

Aiding Intimate Violence Survivors in Lockdown

Lessons about Digital Security in the Covid-19 Pandemic

Julia Slupska

Oxford Internet Institute, University of Oxford
Julia.slupska@stx.ox.ac.uk

Megan Lindsay Brown

Office of Gender-Based Violence, Arizona State University
mlindsa3@asu.edu

ABSTRACT

The global pandemic and associated lockdown measures created new and severe forms of vulnerability for people living in the context of intimate partner violence and coercive control, including being trapped in their homes with abusers and struggling to access advocates as services moved online. This paper outlines innovations used by victim advocates and service providers in reaching survivors in often impossible circumstances, and summarizes lessons from the design of a safety-planning application named “myPlan”. We propose digital security guidelines for products and services which may be used by survivors cohabiting with abusers or using devices which may be under surveillance. We conclude with reflections on whether technology design leads to empowerment: although we cannot overstate the importance of digital security design which is sensitive to the unique vulnerabilities of marginalized people, true empowerment requires a greater commitment to funding accessible housing, mental health support, and advocacy.

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1 INTRODUCTION

The global pandemic and associated lockdown measures created new and severe forms of vulnerability for people living in the context of intimate partner violence (IPV) and coercive control. IPV rates increased significantly as people were confined to their homes, creating a “pandemic within a pandemic” [1]. One recent meta-analysis found an 8.1% increase in domestic violence across the U.S. based on 31 estimates from 12 studies [2]. Further, a systematic review published in 2021 indicates that there was a 16% increase in first time abuse during the pandemic and that increased financial strain caused by the pandemic also led to a significant increase in intimate partner violence, thus exposing socially vulnerable individuals to intimate partner violence in the context of COVID-19 [3]. Although many more IPV victims were now trapped with their abusers, in some regions calls to domestic violence hotlines decreased by as much as 50% [4]. IPV experts worried that this was

because survivors were not able to safely access services with the abuser present. In these circumstances, survivors may be forced to rely on potentially compromised devices to access support, creating “a dangerous dilemma for digital safety” [5].

Although IPV was ignored in most traditional information security threat models [6], technology-enabled abuse has now become a significant area of research [7]–[9]. Research in this field has elaborated the unique challenges of ensuring security in the context of IPV, where the perpetrator may guess or coerce their target’s passwords, surveil messages and devices, and use intimate data to threaten or further coerce the survivor. Technology-enabled abuse further exacerbates the difficulties of providing online mental health support and other services to survivors under lockdown.

Victim advocates and service providers have reacted with a variety of innovations in responding to this difficult problem. This paper reviews existing research in this area, summarizes some lessons from the design of a safety-planning application named “myPlan” (developed by a research team which includes the second author of this paper) and outlines implications for design. We conclude with some reflections on whether technology design can truly lead to empowerment.

2 BACKGROUND AND RELATED WORK

2.1 Technology-enabled IPV

Each year, abuse and coercive control make millions of people vulnerable worldwide, particularly those who are already disadvantaged by discrimination due to gender, race, disability, poverty or other forms of marginalisation [10]–[12]. Previous literature reviews have shown a consistent, and slow stream of qualitative research documenting IPV survivors’ experiences and advocates’ reports of technology-enabled abuse [13], [14]. Existing cybersecurity and human-computer interaction (HCI) research on coercive control has examined survivors’ experiences and security practices [15], [16], perpetrators’ attack strategies [9], [17], [18], and how emerging technologies, such as smart home devices, are co-opted for abuse [19]. Violence researchers have emphasised that seemingly benign tools can be harmful when there is already fear of physical violence [20], [21].

Levy and Schneier [22] define “intimate threats” as privacy threats which arise in intimate relationships and propose design principles to mitigate these; in particular, designers should not assume a household is one unit for the sake of privacy and security design. Resources like the Coercive Control Resistant Design guide provide principles for designers seeking to create technology which is resistant to being used as a tactic in abuse [23]. Cornell’s IPV Tech Research Team started the Clinic to End Tech Abuse (CETA), a consultation service in which trained technologists help survivors

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of abuse assess their devices and discover account compromise, exploitable misconfigurations, and potential spyware [7], [24].

2.2 Aiding Survivors in Lockdown

The covid-19 pandemic provided a powerful tool for perpetrators of IPV to keep their targets socially isolated, scared, and emotionally manipulated [4]. At the same time, many services such as shelters, counselling, and healthcare were impacted negatively or even forced to shut down. Service responders reported workforce strain as a result of moving to video conferencing, and serious problems such as a lack of food or supplies at home and work and housing and financial support for survivors [25].

Many services—for example, Safe Horizon in the US or Refuge in the UK—have turned to chat services as a way to reach survivors who may not be safe speaking on the phone. Many IPV-related websites also provide prominent “escape buttons” which allow survivors to quickly leave a website and erase it from their browser history in case a perpetrator suddenly enters the room.¹ One advocate reported turning to adapting safety-planning to the changed circumstances to focus on emotional safety as well as questions like: “What does it look like for you to create physical distance between yourself and the abuser in the space that you’re in? What does it look like for you to use virtual tools? What does it look like for you to reach out in a time of emergency?” Some courts have set up remote access to file for protection orders against abusers by phone or email [4].

Cornell’s CETA, which previously developed the clinical computer security model for aiding survivors, report on the difficulties in transferring this model online as a result of the pandemic [5]. In addition to traditional security problems facing survivors (i.e. perpetrators may escalate abuse if they find out that survivors reached out to a support service), remote online support also created new risks for providers, namely the potential exposure of their names, faces, or other personal information to abusers who might be listening. To address this, the consultants added additional questions to the consultation whether the client was calling from a safe location on a safe device. This prompted survivors to reflect on the safety of their situation and take additional precautions (such as calling in from a work laptop) if necessary. CETA advocates also implemented measures to preserve anonymity, such as avoiding using names during the consultation. Remote service delivery required substantial extra work and is emotionally taxing in new ways, due to factors such as a lack of in person cues and the potential for a heightened emotional state if the abuser is in the home.

3 LESSONS FROM AN ONLINE IPV INTERVENTION

myPlan² is an online risk-informed safety planning intervention that has been publicly available since 2019. The myPlan app, which was first launched as a website [26], helps individuals experiencing relationship abuse assess their risk for homicide, gain clarity regarding their priorities, and access victim services [27]. The intent of the intervention is to deliver high quality information to those in

an abusive relationship based on several years of violence research [28].

The myPlan app is comprised of three primary intervention components; learn, assess, and safety plan. The user has the ability to move in and out of each section as they choose and each app is linked to a device with a unique pin, essentially a session is ongoing until a user actively logs out in order to begin the safety planning process again. Users can learn more about healthy relationship qualities and spotting abuse in the *learn* section, this portion is purely educational. When a user provides specific information to the app by going through the *assess* section they are receiving valuable feedback (i.e., a real time Danger Assessment, the decision aid) while also providing the app with relationship information (i.e., living with your partner, shared children, abuse history) that is used to populate resources found in the safety plan. The final section of the app is the tailored safety plan, this is a list of local and national resources as well as specific safety tips that can help a survivor manage ongoing abuse and safely leaving if they choose. Below is a description of several studies that have used these three intervention components, providing useful case illustrations and examples of how the intervention is proving effective.

The app has been tested in several intervention studies, including randomized-control trials, quasi-experimental, and pilot testing [29]–[34]. Current available evidence about myPlan shows survivors benefit from the information provided in the app, and there have been no adverse reports from users during the research process. Multiple myPlan studies show survivors reporting reduced decisional conflict; that is, a user gains clarity about their path forward and they experience more confidence in their choices [25]. In research studies across multiple countries (the United States, Canada, New Zealand, and Kenya) findings report that women who used my plan had a greater likelihood of positive safety planning actions [33], [35]. In a study of college-aged women users had significantly less suicidality [35], in a community-based sample the survivors were more likely to report leaving their abusive partner [33], in multiple studies users report they felt more prepared to achieve safety [32], [34], and in a randomized control trial in New Zealand women had reduced symptoms of depression [32]. Cumulatively, this evidence is promising for users and survivor advocates who wish to better understand effective safety planning strategies and especially online safety planning.

In addition to these intervention studies with users there are several published qualitative studies describing the development and testing of the myPlan app and papers regarding the protocol used to establish websites and web content for the studies [36]–[39]. While research has focused on women in relationships with abusive men, content has been adapted to better meet the needs of gender diverse individuals and those in LGBTQ relationships [37]. While other work describes how myPlan functions and general intervention implementation works based on point of contact with a survivor, for instance healthcare workers [40] and pregnant women [41], current content adaptations include a version for youths, Indigenous women, and immigrant women in the U.S. [36], [39]. It is noteworthy that the app is being thoughtfully tested based on population need and survivor service settings to ensure users can receive the most tailored version possible for their unique needs.

¹See, for example, <https://www.womensaid.org.uk/>

²www.myplanapp.org

Serious care and consideration must be taken when launching a product meant to be visible and understood by only some potential users (i.e. survivors but not perpetrators). At the design and implementation stages of the myPlan app, several additional steps were taken to protect users:

- Domestic violence colors (i.e., purple the domestic violence campaign colour in the United States) and traditional branding was not included.
- The app requires users to create a pin-code. An additional dummy pin-code can be used in order to redirect to an unrelated website if an abuser is going through the phone.
- The app and website do not require a login and no personal identifying information is needed to create a safety plan. The site information is connected to the device rather than an individual.
- There are quick-exit features.
- Users data is not stored, analyzed, or used for any form of research or marketing. All research studies using the intervention operate from a different site than the public version of myPlan.
- The social media pages focus on relationship health and do mention domestic violence, but are intended to attract a broad audience of potential users.

In addition to these design considerations, over the past three years the myPlan team trained faculty, students, staff and local victim advocates working in domestic violence programs. Live presentations and local study implementation created an informal, but useful channel of feedback from users as they tested the app in live demonstrations. Unexpected feedback included adopting the app as a trauma informed practice and way for advocates and survivors to communicate and the appropriateness of using mobile safety plans given the high rate of women leaving shelter settings unexpectedly. As users tried the app in demonstrations the need for a high-functioning well designed platform proved to be as powerful, if not more so, than the quality of information. Repeatedly advocates emphasized the quality of technological functioning influences their own, and their clients, ability to trust a resource as legitimate. Digital interventions are particularly useful as many may not be able to access in person services or use calls or audio during lockdowns, yet we need a greater investment from public funding sources, private industry, and non-profit groups to ensure these tools remain high quality and sustainable over time.

4 IMPLICATIONS FOR DIGITAL SECURITY DESIGN

Anyone creating digital services which are aimed at, or may be used by, survivors of IPV should keep in mind the new vulnerabilities created by the covid-19 pandemic and associated lockdowns. This is particularly relevant for victim advocates or those providing online mental or physical health services but could also apply to a wide range of government services which have moved online, online social networks, or health trackers. In this section, we summarise crucial digital security considerations based on our review of existing literature and lessons from myPlan. To derive these implications, we drew on the phenomenon of technology-mediated abuse as a form of “sensitizing concept” which highlights relevant

social aspects concerning technology use and social problems [42]. Sensitizing concepts allow design researchers to state functionalities that a class of systems should have. Furthermore, in looking at successful interventions, such as Cornell’s CETA [5] and myPlan [26] allows us both to draw on previous fieldwork-informed design implications and to critically analyze successful systems to try to replicate their successes using related design concepts [29]. Therefore design implications were chosen on the basis of previous cases where they had been successful, and on the principle of actionability, ie something that can feasibly be implemented by technologists.

4.1 Risk assessment

Practitioners are usually trained in the use of risk assessments to predict physical danger for an IPV survivor, however these assessments should also include the possibility of compromised devices. Asking questions about whether the survivors is using a safe device in a safe place offers survivors a prompt to reflect on potential implications of using the device for services, as well as possibilities to do so more safely [5].

4.2 Escape functions and access controls

Sites or products which offer information or content which might expose a survivor to further risk if seen by a perpetrator (for example, information about domestic violence services or LGBTQ+ content) should offer easy, prominent “escape” or “exit site” button to account for the risk of an abuser suddenly entering the room. Apps prompting for sensitive information should use pin codes and offer a dummy pin-code to redirect the page and hide sensitive content from an abuser.

4.3 Plurality of communication methods

Offering a plurality of communication methods: such as text, email, or audio gives survivors more options to choose a method which is safe and comfortable for them. It is critical to keep in mind that survivors may experience a traumatic reaction in response to certain digital contexts; for example, if a survivor was stalked or harassed on Facebook they may avoid the site and will not be able to access information or services provided there. Our work in shelters and sexual abuse listening services confirms the need for this; device access and experience varies widely and some survivors are much more responsive via text.

4.4 Accessible security advice

Victim advocates have an informal mantra: *survivors have the most expertise about their own abuse*. To best serve a survivor of violence practitioners are trained first to believe, and second to navigate the situation with trust regarding the survivor’s perceptions about response. Security advice should be delivered in a manner that acknowledges the level of harm, but does not intimidate or confuse survivors and the community practitioners providing services. Violence researchers have been exploring the best safety strategies, and recent research explores how digital security advice can incorporate feminist principles to increase accessibility by avoiding victim-blaming and jargon technical language [43]. Some practitioners also suggest neutral terms like “context analysis” instead of

terms like “risk assessment” or “threat modelling” which might be stressful or intimidating [44]. Websites and apps providing services to survivors should seek to keep them informed about security practices which may help them stay safe, such as using safe passwords or deleting potentially compromising sites from browser history, and train victim advocates to assist so the survivors can focus on other significant components of personal safety.

4.5 Secure & de-identified storage

Increasingly, survivors must collect, manage and store data as evidence in order to demonstrate the level of abuse and justification for services. Shifting toward a model where survivors must document their own abuse should give advocates and policy makers pause as it minimizes the risk survivors are often taking by implementing such a practice. Several research studies have found abusive partners regularly check phones, both overtly and covertly, and storing sensitive information on their own device could trigger additional violence [18], [45], [46]. Further, research indicates when abuse has begun escalating (increased frequency, severity) this can portend intimate partner homicide [47] further justifying caution for documentation, and more robust supports for technology safety as survivors are seeking help.

5 DISCUSSION

Technology can enable survivors to make more informed choices and develop their own capacities, however the language of “empowerment” can be misleading in this context. As Ereteet et al. [28] write, technology interventions alone cannot empower people to solve social problems without addressing underlying social, economics, and political inequities. Survivors are often targeted because they belong to a marginalized group and abusers know they can wield systemic power against them. To describe projects and technologies as empowering when they do not truly shift power can obscure this reality [5].

As researchers on the myPlan team often repeated; this is an intervention and not an app. The digital intervention should never be used to replace, but rather to supplement the work of advocates. Women stay in abusive relationships for multiple reasons including financial dependence, the safety of their children, and sometimes as a strategy for personal safety [48], [49]. The most effective treatment for IPV survivors is not additional information, but connection with a trusted practitioner. At best the app can organize and simplify information, but it cannot provide emotional and caring support in order for the survivor to begin trusting their own intuition to leave the relationship and seek out help. Therefore, greater funding for housing, healthcare, and mental health in victim advocacy—an often critically under-funded field—is crucial to truly address these problems. For example, the non-profit and research sectors have yet to find a sustainable pathway to financial support for myPlan itself.

Similarly, the design guidelines we advocate for in this paper are better understood as a form of harm reduction than empowerment. Failing to take into account the unique and challenging vulnerabilities faced by survivors under lockdown would expose them to greater harm.

6 CONCLUSION

In this paper we outline new forms of vulnerability experienced by survivors of IPV during covid-19 lockdowns as well as how digital security measures can support survivors in often impossible circumstances. Drawing on lessons from existing interventions, such as the Clinic to End Tech Abuse as well as the myPlan app, we propose several design guidelines which designers and service providers should keep in mind when creating services or products which may be used by survivors co-habiting with an abuser. We conclude with some reflections on the notion of technology design as a lead to empowerment. Although we cannot overstate the importance of digital security design which is sensitive to the unique vulnerabilities of marginalized people, true empowerment requires a greater commitment to accessible housing, mental health support, and advocacy. To quote Safiya Noble, “An app will not save us” [50].

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